

Autosomal Dominant Hearing Loss 78 (DFNA78, SLC12A2-related) — Comprehensive Disease Report

Prepared as a disease knowledge-base entry. Evidence types are flagged as [human clinical], [model organism], [in vitro], or [computational]. Primary citations are given as PMIDs.

Summary / Consensus Answer

DFNA78 is a rare, autosomal dominant, non-syndromic sensorineural hearing loss (frequently accompanied by bilateral vestibular areflexia) caused by heterozygous variants in **SLC12A2**, the gene encoding the secretory $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter **NKCC1**. The pathogenic variants cluster in **exon 21 (or its 3' splice site)**, a region present almost exclusively in the cochlea-specific NKCC1 isoform that is required to maintain the K^+ -rich endolymph and endocochlear potential that drive hair-cell mechanotransduction. Reduced/dysfunctional cotransport disrupts endolymph homeostasis, leading to hair-cell dysfunction and hearing/vestibular loss. Management is supportive (hearing aids, cochlear implantation, vestibular rehabilitation); no disease-specific pharmacotherapy exists.

1. Disease Information

Overview. DFNA78 ("Deafness, autosomal dominant 78") is a Mendelian sensorineural hearing loss (SNHL) caused by heterozygous *SLC12A2* variants. It was first delineated in 2020 [human clinical;

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32658972]. The phenotype is congenital-to-early-onset, bilateral SNHL, often severe-to-profound, and frequently with **bilateral vestibular areflexia** (hence sometimes described as a cochleovestibular defect) [PMID 32658972; 40503591].

"In 2020, heterozygous variants in SLC12A2 were identified as a cause of non-syndromic deafness associated with vestibular areflexia (DFNA78; MIM 619081)." —

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Key identifiers (verified via OLS4/MONDO, 2026-09-03). - **OMIM:** #619081 (Deafness, autosomal dominant 78) - **Gene OMIM:** SLC12A2 600840 - **MONDO:** MONDO:0033665 ("hearing loss, autosomal dominant 78") - **DOID:** DOID:0112159; **GARD:** 0018156; **MedGen:** C5436768 (CUI 1777362); **UMLS:** C5436768 - **Orphanet:** *No dedicated DFNA78 ORPHAcode* (MONDO records no Orphanet xref); clinically grouped under autosomal dominant non-syndromic sensorineural hearing loss. - **ICD-11:** AB52 Sensorineural hearing impairment (bilateral, AB52.0); **ICD-10:** H90.3/H90.5 - **MeSH:** "Hearing Loss, Sensorineural" (D006319); "Hearing Loss, Bilateral" - **HGNC gene:** HGNC:10911 (SLC12A2); **NCBI Gene:** 6558; **Ensembl:** ENSG00000064651; **UniProt:** P55011 (NKCC1)

Synonyms / alternative names: DFNA78; deafness, autosomal dominant 78; SLC12A2-related autosomal dominant deafness; NKCC1-related hearing loss; non-syndromic sensorineural hearing loss with vestibular areflexia. **Distinct allelic disorders** (same gene, different phenotype/inheritance): **Delpire-McNeill syndrome** — neurodevelopmental disorder, MONDO:0033667, OMIM #619083, Orphanet:633024 [PMID 32658972]; **Kilquist syndrome** — autosomal recessive multisystem disorder, MONDO:0033664, OMIM #619080, Orphanet:633021 [PMID 30740830].

Data source type: Aggregated disease-level knowledge derived from a small number of published case reports/families and functional studies (not EHR-derived at scale). Fewer than ~20 unrelated affected individuals/families are described to date.

2. Etiology

Primary cause — genetic. DFNA78 is a monogenic disorder caused by heterozygous, usually dominantly-inherited or de novo, variants in *SLC12A2* [human clinical;

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32658972]. There is **no environmental or infectious cause**; environmental agents are irrelevant to the primary etiology.

Genetic risk factors. - Causal variants: missense and splice-site variants restricted to **exon 21 / its 3' splice site** of the cochlea-specific NKCC1 isoform [PMID 32294086]. ClinVar additionally lists DFNA78-annotated truncating alleles **c.2977G>T (p.Glu993Ter)** [Pathogenic] and **c.869dup (p.Val291fs)** [Likely pathogenic] [computational; ClinVar 2026]. - No known common susceptibility loci or modifier genes have been established for DFNA78 (the disorder is monogenic and ultra-rare).

Environmental risk / protective factors. None established as causal or modifying for DFNA78. General SNHL aggravators (noise, ototoxic aminoglycosides, loop diuretics) are theoretical concerns because loop diuretics (bumetanide/furosemide) inhibit NKCC1; avoidance is prudent but not evidence-based for this specific disease.

Gene–environment interactions. Not characterized. Given the ion-transport mechanism, ototoxic drugs that further impair endolymph homeostasis (loop diuretics, aminoglycosides, cisplatin) could plausibly worsen hearing, but no formal GxE data exist.

3. Phenotypes

Core phenotype = **bilateral sensorineural hearing loss ± bilateral vestibular areflexia**, non-syndromic.

Phenotype	Type	Onset	Severity	Progression	Frequency	HPO term
Sensorineural hearing loss	Clinical sign / lab (audiometry)	Congenital–early childhood (some later/progressive)	Moderate → profound	Stable or slowly progressive	~100% (defining)	HP:0000407 Sensorineural hearing impairment; HP:0008619 Bilateral SNHL; HP:0008527 Congenital SNHL; HP:0000408 Progressive SNHL
Vestibular areflexia / dysfunction	Clinical sign (caloric/vHIT)	Congenital–early	Variable	Stable	Frequent (subset; reported in the cochleovestibular families)	HP:0410057 Vestibular areflexia; HP:0002321 Vertigo; HP:0001336 (imbalance)
Delayed motor milestones (2° to vestibular loss)	Physical	Infancy	Mild	Non-progressive	Occasional	HP:0001270
Tinnitus	Symptom	Variable	Mild-moderate	Variable	Not systematically reported	HP:0000360

Explicitly NOT part of DFNA78 (these distinguish it from the recessive/de-novo allelic disorders): intellectual disability, developmental delay, gastrointestinal dysmotility, absent salivation/lacrimation, respiratory abnormalities, microcephaly — these occur in **Kilquist syndrome** (biallelic LoF) and **Delpire-McNeill** (de novo NDD) [PMID 30740830; 32658972; 33500540].

Quality-of-life impact. SNHL impairs speech/language acquisition, education, and communication; vestibular areflexia impairs balance, gaze stabilization (oscillopsia), and delays motor development. No DFNA78-specific EQ-5D/SF-36 data; QoL burden inferred from congenital bilateral SNHL literature generally.

4. Genetic / Molecular Information

Causal gene. *SLC12A2* (Solute carrier family 12 member 2), encoding **NKCC1** (Na⁺-K⁺-2Cl⁻ cotransporter 1). HGNC:10911; NCBI Gene 6558; Ensembl ENSG00000064651; UniProt **P55011** (1212 aa); locus **5q23.3** (GRCh38 chr5:128,083,766–128,189,677); canonical transcript ENST00000262461 / NM_001046.3 [computational; gnomAD/UniProt].

Pathogenic variants. - **Type/class:** predominantly **missense** and **splice-site** variants in exon 21 / its 3' splice acceptor (cochlear isoform) [PMID 32294086]; ClinVar also records truncating (nonsense/frameshift) DFNA78 alleles (p.Glu993Ter; p.Val291fs) [ClinVar]. - **Classification (ACMG/AMP):** the two established DFNA78 alleles are Pathogenic / Likely pathogenic; most other *SLC12A2* variants are VUS [ClinVar 2026]. - **Allele frequency:** private/ultra-rare; absent or singleton in gnomAD (consistent with de novo or small-family dominant transmission). - **Origin: germline** — inherited (dominant families) or **de novo** (sporadic cases) [PMID 32294086; 32658972]. No somatic involvement. - **Functional consequence:** reduced cotransporter activity — "Cl⁻ influx was significantly decreased in all SLC12A2 variants studied" [in vitro Xenopus oocytes;].

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Because NKCC1 is an obligate **dimer** and heterozygous LoF carriers have normal hearing, DFNA78 missense alleles likely act via **dominant-negative** / **isoform-specific** loss of endolymph-secreting function rather than simple haploinsufficiency (partly inferred) [PMID 40503591].

"SLC12A2 functions as a dimer and has several isoforms; only one isoform contains exon 21, and this isoform is almost exclusively expressed in the inner ear/cochlea." —

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Constraint (gnomAD). LOEUF 0.60, LoF o/e 0.49 (obs 67/exp 136), lof_z 5.04, **pLI $\approx$ 0**, missense o/e 0.86 (mis_z 2.32) — i.e., *SLC12A2* is only moderately LoF-constrained and **not classically haploinsufficient**, supporting the non-haploinsufficiency mechanism [computational; gnomAD].

Modifier genes / epigenetics / chromosomal abnormalities. None established for DFNA78. (In Kilquist syndrome the causal lesion was a homozygous 22-kb deletion arising via **uniparental isodisomy of chromosome 5** — a large-scale mechanism relevant to the recessive allelic disorder, not DFNA78 [PMID 30740830].)

5. Environmental Information

Not applicable to disease causation. No environmental toxins, lifestyle factors, or infectious agents are implicated in DFNA78. Theoretical aggravators of any SNHL — noise exposure, aminoglycosides, cisplatin, and **loop diuretics (NKCC1 inhibitors)** — should be minimized as general otoprotective practice, but there is no disease-specific evidence.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A heterozygous *SLC12A2* variant in the cochlea-specific **exon-21 region** (missense/splice/truncating) **alters NKCC1 in the exon-21 (inner-ear) isoform** [PMID 32294086].
2. The mutant subunit **incorporates into the obligate NKCC1 dimer**, and (inferred) **exerts a dominant-negative / isoform-specific reduction** of cotransport in cochlear secretory epithelium — demonstrated as decreased Cl^- influx in vitro [PMID 32294086; dimer/mechanism partly inferred,

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3. Reduced NKCC1 activity in **stria vascularis marginal cells** (and other lateral-wall cells) **decreases basolateral uptake of $\text{Na}^+/\text{K}^+/\text{Cl}^-$** , which **results in reduced K^+ secretion into the endolymph** [model organism;

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4. Impaired K⁺ secretion **leads to failure of endolymph homeostasis and reduction of the endocochlear potential (EP)** — the driving force for hair-cell transduction currents [PMID 32294086; 17674100].
5. Loss of the K⁺ gradient/EP **results in failure of hair-cell mechano-electrical transduction** ("K⁺ transport is required for the mechano-transduction of auditory stimuli") [PMID 40503591].
6. Chronic endolymph disturbance **leads to structural damage / collapse of the endolymphatic compartment and hair-cell dysfunction** (shown in Slc12a2^{-/-} mice) [model organism; P 10369265].
7. → **Sensorineural hearing loss**; in parallel, the same defect in **vestibular dark cells** of the labyrinth **results in vestibular hypofunction/areflexia** [PMID 32658972].

Molecular pathways / biochemistry. The core defect is an **ion-transport (ion-homeostasis) defect**, not a classical signaling cascade. NKCC1 mediates electroneutral, bumetanide-sensitive Na⁺:K⁺:2Cl⁻ symport (MF **GO:0008511** sodium:potassium:chloride symporter activity). Relevant GO biological processes (UniProt P55011): potassium ion transmembrane transport (**GO:0071805**), chloride transmembrane transport (**GO:1902476**), potassium ion import across plasma membrane (GO:1990573), intracellular potassium ion homeostasis (GO:0030007), cell volume homeostasis (GO:0006884), sensory perception of sound (**GO:0007605**), inner ear morphogenesis (GO:0042472). CHEBI entities: potassium(1+) **CHEBI:29103**, chloride **CHEBI:17996**, sodium(1+) CHEBI:29101; inhibitors bumetanide **CHEBI:3213**, furosemide CHEBI:47426.

Protein dysfunction. NKCC1 is a 12-TM cation-chloride cotransporter that assembles as a homodimer; cryo-EM structures reveal ion-binding sites and the dimer interface [in vitro/structural; P 36239040];

32081947; 37545407]. Exon-21 variants perturb the C-terminal cytoplasmic domain critical for transport/regulation; recessive alleles also cluster in the C-terminal domain [PMID 33500540]. Consequence = loss/reduction of transport function (± dominant-negative on the dimer).

Cell types / processes (upstream→downstream). Upstream: dysfunction of **strial marginal cells (CL:0002516)** and vestibular dark cells (epithelial K⁺ secretion). Downstream: secondary dysfunction/degeneration of **cochlear hair cells (inner CL:0000589; outer CL:0000601)**,

vestibular hair cells (CL:0000633), and (later) spiral ganglion neurons (CL:0002253). No inflammation/autoimmunity, apoptosis-driven primary mechanism, or metabolic disorder is implicated in DFNA78.

Molecular profiling. NKCC1 immunolocalizes to the basolateral membrane of stria marginal cells; in hereditary-deaf animal cochleae, spatiotemporal loss of SLC12A2 accompanies stria malformation and endolymph collapse [PMID 18093167; 24713161]. No DFNA78-specific transcriptomic/proteomic/metabolomic datasets exist.

7. Anatomical Structures Affected

- **Organ level (primary): inner ear / cochlea** (UBERON:0001846 inner ear; **UBERON:0001844 cochlea**); vestibular apparatus (semicircular canals UBERON:0001840, otolith organs).
 - **Body systems:** special sense (auditory/vestibular) organ of the **nervous/sensory system**. No cardiovascular, GI, renal, or respiratory involvement in DFNA78 (unlike the multisystem recessive disorder).
 - **Tissue/cell level:** cochlear **lateral wall / stria vascularis** (UBERON:0002429) secretory epithelium — **stria marginal cells** (CL:0002516); **organ of Corti** (UBERON:0002227) — inner/outer hair cells (CL:0000589 / CL:0000601); vestibular dark cells and vestibular hair cells (CL:0000633); endolymph compartment (UBERON:0011078).
 - **Subcellular level:** **basolateral plasma membrane** (GO:0016323) of secretory epithelial cells (NKCC1 also at apical membrane GO:0016324 in some epithelia); integral membrane transporter.
 - **Localization / laterality:** **bilateral** (typically symmetric) cochleovestibular involvement.
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8. Temporal Development

- **Onset:** congenital or early-childhood in most reported cases; some heterozygous families show later-onset/progressive SNHL [PMID 32294086; 32658972]. Pattern is **chronic/insidious** rather than acute.
- **Progression:** ranges from **stable** to **slowly progressive**; severity moderate-to-profound. Vestibular areflexia is generally stable/non-progressive.

- **Course/duration: lifelong, non-remitting.** No spontaneous remission; no relapsing-remitting pattern.
 - **Critical period:** early identification (newborn hearing screening) and early amplification/cochlear implantation within the first years of life are the key windows for language outcomes.
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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (DFNA78), including **de novo** occurrences in sporadic cases [PMID 32294086; 32658972].
 - **Penetrance:** appears high in reported dominant families (5/5 affected members in the index family) but formal penetrance estimates are unavailable given few families; treat as **likely high but incompletely quantified**.
 - **Expressivity: variable** (severity and presence of vestibular involvement differ across individuals).
 - **Anticipation / germline mosaicism / founder effects:** none reported. Founder alleles not described.
 - **Carrier frequency:** not applicable (dominant); *SLC12A2* LoF alleles exist in gnomAD but do not cause DFNA78.
 - **Epidemiology: ultra-rare;** exact prevalence/incidence unknown. DFNA (autosomal dominant non-syndromic) deafness collectively is a minority of hereditary hearing loss; DFNA78 is among the rarer subtypes with <20 reported families/individuals. No prevalence per 100,000 can be reliably stated.
 - **Demographics:** reported across multiple populations (Japanese index families [PMID 32294086]; European/other cohorts [PMID 32658972; 40503591]). **No sex predilection** expected (autosomal). No specific geographic/ethnic clustering established.
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10. Diagnostics

Clinical/functional tests. - **Audiometry** (pure-tone, ABR/auditory brainstem response, OAE, tympanometry): documents bilateral SNHL; OAEs typically absent. LOINC panels for audiometry. - **Vestibular testing:** video head-impulse test (vHIT), caloric testing, VEMP,

rotational chair — reveal vestibular areflexia/hypofunction. - **Imaging:** MRI/CT of temporal bones — usually **normal inner-ear anatomy** (helps exclude structural/EVA causes); no pathognomonic finding. - **No blood/urine biomarker** exists; diagnosis is not made by chemistry.

Genetic testing (diagnostic gold standard). - Recommended approach: **hereditary-hearing-loss gene panel** or **whole-exome sequencing (WES)**, with attention to *SLC12A2* including **exon 21 and splice sites** (which may be under-covered/mis-annotated because the pathogenic cochlear isoform differs from the canonical transcript) [PMID 32294086]. WES/WGS with trio analysis is especially useful to detect **de novo** variants [PMID 32658972]. - Single-gene *SLC12A2* testing is appropriate when the phenotype (congenital SNHL + vestibular areflexia, dominant/de novo) is suggestive. - CMA/karyotype/FISH generally not indicated (point-variant disorder); mtDNA and repeat-expansion testing not relevant. - **Variant-interpretation caveat:** confirm variant is in the exon-21 cochlear isoform and, ideally, functional confirmation (reduced Cl⁻ transport) supports pathogenicity [PMID 32294086].

Clinical criteria / differential diagnosis. No formal diagnostic criteria. Differential includes other non-syndromic SNHL genes (*GJB2*, *SLC26A4*/Pendred, *MYO7A*, *TMC1*, etc.), and syndromic causes with vestibular involvement (Usher syndrome). The **combination of non-syndromic congenital SNHL with bilateral vestibular areflexia and dominant/de novo inheritance** points toward *SLC12A2*. Distinguish from the allelic **Kilquist** (recessive, multisystem) and **Delpire-McNeill** (NDD) disorders.

Screening. Detected via **universal newborn hearing screening** (phenotype), then genetic confirmation; **cascade testing** of at-risk relatives in dominant families.

11. Outcome / Prognosis

- **Survival/mortality:** DFNA78 is **not life-limiting**; normal life expectancy. Mortality not attributable to the disease. (Contrast with severe multisystem Kilquist syndrome.)
- **Morbidity/disability:** principal morbidity is **communication disability** from bilateral SNHL and **balance impairment/oscillopsia** from vestibular areflexia; motor milestone delay possible in infancy.
- **Disease course:** chronic, lifelong; hearing stable-to-progressive.

- **Recovery potential:** no spontaneous recovery; **functional hearing is recoverable with amplification/cochlear implantation**, and balance improves with vestibular rehabilitation/compensation.
- **Prognostic factors:** degree/onset of hearing loss, timing of intervention (early implantation → better language outcomes), and residual hearing. No molecular prognostic biomarker established.
- **QoL measures:** no DFNA78-specific instruments; general pediatric SNHL QoL tools apply.

12. Treatment

No disease-modifying/curative pharmacotherapy exists. Management is **supportive/rehabilitative**.

- **Amplification / devices: hearing aids** for milder loss; **cochlear implantation** for severe-to-profound loss (NCIT: Cochlear Implant C99286; Hearing Aid C99285). Cochlear implants bypass the failed endolymph-dependent transduction by directly stimulating the auditory nerve and are expected to be effective (spiral ganglion typically preserved early).
 - **Vestibular rehabilitation** / physical therapy for balance; **speech-language therapy** and early-intervention/educational support.
 - **Pharmacotherapy:** none specific. Note NKCC1 is the target of **loop diuretics (bumetanide CHEBI:3213, furosemide)** — these *inhibit* NKCC1 and are potentially ototoxic, so they are **not therapeutic** and are best avoided. Bumetanide is being studied in NKCC1-related neurodevelopmental/GABA-polarity contexts [PMID 26955005; 38950809], but this is **not applicable to restoring cochlear NKCC1 function** in DFNA78.
 - **Advanced/experimental:** No approved gene therapy, ASO, or cell therapy for DFNA78. Given the dominant/likely dominant-negative mechanism, allele-selective silencing (siRNA/ASO) or gene editing are conceptually attractive future strategies but remain **preclinical/theoretical**. No registered clinical trials (ClinicalTrials.gov) target DFNA78 specifically as of this report.
 - **Pharmacogenomics:** avoid concomitant ototoxic drugs (aminoglycosides, cisplatin, loop diuretics).
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13. Prevention

- **Primary prevention:** not possible (genetic). **Genetic counseling** for dominant families (50% transmission risk); options include **prenatal testing** and **preimplantation genetic testing (PGT)** for known familial variants.
 - **Secondary prevention / early detection:** **universal newborn hearing screening** enables early diagnosis and intervention; **cascade genetic testing** of relatives.
 - **Tertiary prevention:** early amplification/cochlear implantation and vestibular rehabilitation to prevent language/developmental sequelae; avoidance of ototoxic exposures and noise to preserve residual hearing.
 - **Immunization / public-health / environmental measures:** not applicable.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *Slc12a2* is highly conserved. Mouse **Slc12a2** (NCBI Gene 20496; MGI:101924; NCBI Taxon 10090); rat *Slc12a2* (Taxon 10116); zebrafish *slc12a2* (Taxon 7955). Human ortholog UniProt P55011.
 - **Natural disease in animals:** Deafness/inner-ear dysfunction from *Slc12a2* disruption is a **model-organism phenomenon** (targeted mutants); no well-known spontaneous companion-animal DFNA78 equivalent is catalogued (OMIA lists NKCC1/SLC12A2 biology but no established dominant deafness breed disorder).
 - **Comparative biology:** NKCC1's role in endolymph/K⁺ cycling is conserved across mammals (mouse, gerbil, rat, guinea pig) and underlies the shared deafness phenotype [PMID 10369265; 18093167]. Evolutionary conservation of the endolymph-secretion mechanism is strong.
 - **Transmission:** none (non-infectious, non-zoonotic).
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15. Model Organisms

- **Mouse — *Slc12a2* (NKCC1) knockout** [model organism;

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constitutive KO mice are **deaf**, display classic **shaker/waltzer** circling/head-bobbing (vestibular dysfunction), and show **collapse of the endolymphatic compartment** and stria/inner-ear structural damage from reduced endolymph secretion. This recapitulates the core DFNA78 mechanism (endolymph/EP failure) and the cochleovestibular phenotype.

"absence of functional co-transporter leads to structural damages in the inner ear consistent with a decrease in endolymph" —

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- **Allelic mouse models / EP studies:** heterozygous *Nkcc1* deletion causes **progressive, age-dependent hearing loss** with EP decline; combined manipulation with Na,K-ATPase isoforms modulates EP and hearing, underscoring NKCC1's role in EP generation [PMID 17674100].
- **Other species:** guinea-pig hereditary-deafness models show spatiotemporal loss of stria K^+ -transport proteins including SLC12A2 with endolymph collapse [PMID 18093167].
- **Model types available:** constitutive knockout; heterozygous; the biology also supports conditional/knock-in and humanized approaches (MGI/IMPC resources for *Slc12a2*).
- **Recapitulation & limitations:** KO models faithfully reproduce **deafness + vestibular dysfunction + endolymph collapse**, but represent **complete loss of function/biallelic** biology (closer to Kilquist) rather than the human **heterozygous, isoform-/exon-21-specific dominant** mechanism; a precise DFNA78 exon-21 knock-in would better model the dominant-negative disease. No DFNA78 iPSC/organoid model is yet published.
- **Databases:** MGI (*Slc12a2*), IMPC/IMSR, Alliance of Genome Resources, ZFIN (*slc12a2*), RGD.

Supported vs Refuted Hypotheses

Supported: - DFNA78 is caused by heterozygous *SLC12A2* variants clustered in the cochlea-specific exon-21 region [PMID 32294086]. - Mechanism = impaired NKCC1-mediated K^+ secretion → endolymph/EP failure → hair-cell transduction failure → SNHL + vestibular areflexia [PMID 10369265; 32294086; 40503591]. - *SLC12A2* is an allelic series (dominant DFNA78 vs recessive Kilquist vs de novo Delpire-McNeill NDD) [PMID 30740830; 32658972; 33500540].

Refuted / disfavored: - **Haploinsufficiency as the DFNA78 mechanism** — disfavored: gnomAD shows *SLC12A2* is not classically haploinsufficient ($pLI \approx 0$) and heterozygous LoF carriers/patients have normal hearing; DFNA78 arises from isoform-specific missense/splice (dominant-negative) alleles [PMID 40503591; gnomAD]. - Environmental/infectious causation — not applicable.

Limitations & Future Directions

- Very small number of reported families → penetrance, expressivity, prevalence, natural-history, and audiometric-progression data are limited.
 - The dominant-negative mechanism is **inferred** from the dimeric structure and genotype-phenotype correlation; direct in-vivo proof (exon-21 knock-in models) is lacking.
 - No DFNA78-specific omics, iPSC/organoid, or therapeutic-trial data. Future work: exon-21 knock-in mouse, patient iPSC-derived otic organoids, and allele-selective RNA/gene-editing therapeutics.
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Key References (PMIDs)

- 32294086 — Mutai et al. 2020: exon-21 *SLC12A2* variants cause hereditary hearing loss (foundational DFNA78).
- 32658972 — McNeill et al. 2020: *SLC12A2* variants cause NDD or cochleovestibular defect (vestibular areflexia).
- 40503591 — Ludin et al. 2025: de novo *SLC12A2* variant, congenital HL + vestibular areflexia; isoform/exon-21 review.
- 30740830 — Macnamara et al. 2019: Kilquist syndrome (biallelic *SLC12A2* deletion).
- 33500540 — Bilal Shamsi et al. 2021: autosomal recessive *SLC12A2* disease; C-terminal clustering.
- 10369265 — Delpire et al. 1999: *Slc12a2*-KO mouse deafness/imbalance, endolymph collapse.
- 17674100 — Diaz et al. 2007: NKCC1/Na,K-ATPase, endocochlear potential and hearing.
- 18093167 — Jin et al. 2008: loss of strial K^+ -transport proteins (incl. *SLC12A2*) in hereditary deafness.
- 36239040 / 32081947 / 37545407 — human NKCC1 cryo-EM structures.
- 33345190 / 37399495 — reviews: NKCC1 as a human disease-causing ion transporter.

- 26955005 / 38950809 — SLC12A2/NKCC1 GABA-polarity and bumetanide (neuro contexts; not DFNA78 therapy).

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